Uniform plane wave propagation

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$$\nabla^2 \vec{E} = -k^2 \vec{E}$$

Similarly we can derive $\nabla^2 \vec{H} = -k^2 \vec{H}$ } vector wave equations

"Helmholtz wave equations"

$$jk = \sqrt{-\omega^2 \mu (\epsilon' - j\epsilon'')}$$

Using the calculus one can write it as $jk = \alpha + j\beta$

where

$$\alpha = \sqrt{\frac{\omega^2 \mu \epsilon'}{2}\left[\left(1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}} - 1\right]}$$

and

$$\beta = \sqrt{\frac{\omega^2 \mu \epsilon'}{2}\left[\left(1 + \left(\frac{\epsilon''}{\epsilon'}\right)^2\right)^{\frac{1}{2}} + 1\right]}$$

$$\epsilon' = \epsilon_0 \epsilon_r$$

$$\epsilon'' = \frac{\sigma}{\omega}$$

Wave equation

Let us simplify further

Let $\vec{E}(x,y,z)e^{j\omega t} = \vec{E}(z)e^{j\omega t}$ → wave in z direction only

$= \hat{a}_E E(z) e^{j\omega t}$

Substitute it in vector wave equation $\nabla^2 \vec{E} = -k^2 \vec{E}$

↳ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)$

$$\frac{\partial}{\partial z^2} \hat{a}_E E(z) \cdot e^{j\omega t} = -k^2 \hat{a}_E E(z) \cdot e^{j\omega t}$$

But E is function of z only

Removing common terms

$$\frac{\partial}{\partial z^2} E(z) = -k^2 E(z)$$

Solve this differential equation (Integrate twice & find constants)

$$\boxed{E(z) = E_0 e^{\pm jkz}}$$

Wave equation

$$\vec{E}(x,y,z) = \hat{a}_E E(z) e^{j\omega t}$$

$$= \hat{a}_E E_0 e^{\pm jkz} \cdot e^{j\omega t}$$

$$= \hat{a}_E E_0 e^{j(\omega t \pm kz)}$$

The real part of this is

$$\vec{E}(x,y,z) = \hat{a}_E E_0 \cos(\omega t \pm kz)$$

This is the wave equation we used to represent uniform plane wave in previous class

Use either $\hat{a}_x$ or $\hat{a}_y$

Intrinsic impedance

Let $\vec{E} = \hat{a}_x E_0 e^{j(\omega t \pm kz)}$

Take curl $\rightarrow \nabla \times \vec{E} = -j\omega\mu\vec{H}$

$$\nabla \times \vec{E} = \begin{vmatrix} a_x & a_y & a_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_0 e^{j(\omega t \pm kz)} & 0 & 0 \end{vmatrix} = -j\omega\mu\vec{H}$$

$$\Rightarrow \pm \hat{a}_y jk E_0 e^{j(\omega t \pm kz)} = -j\omega\mu H$$

$$\Rightarrow H = \frac{\mp \hat{a}_y jk}{j\omega\mu} E_0 e^{j(\omega t \pm kz)}$$ This must be equal to $\hat{a}_y H_0 e^{j(\omega t \pm kz)}$

$$\Rightarrow H_0 = \frac{k}{\omega\mu} E_0 \Rightarrow \eta = \frac{E_x}{H_y} = \frac{\omega\mu}{k} = \frac{j\omega\mu}{\alpha + j\beta}$$


Thank
you